

Lesson 4

The Frisch–Waugh–Lovell theorem, omitted and irrelevant variables, and the linear hypothesis

1 The model, in two blocks (a reminder)

The set opens with the model of the previous lessons written again,

$$y = X\beta + u,$$

and then split into two groups of regressors:

$$y = \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + u = X_1\beta_1 + X_2\beta_2 + u,$$

with the dimensions read under the terms of the line,

$$\underbrace{y}_{n \times 1} = \underbrace{X_1}_{n \times k_1} \underbrace{\beta_1}_{k_1 \times 1} + \underbrace{X_2}_{n \times k_2} \underbrace{\beta_2}_{k_2 \times 1} + \underbrace{u}_{n \times 1}.$$

The estimator is $b = (X'X)^{-1}X'y$, and written out in the same two blocks,

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{bmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{bmatrix}^{-1} \begin{pmatrix} X_1'y \\ X_2'y \end{pmatrix}.$$

2 The projection and the annihilator, and the two block pairs

The reminder continues with the two matrices,

$$P = X(X'X)^{-1}X' \implies PX = X \implies Pe = 0, \quad M = I_n - X(X'X)^{-1}X' \implies MX = 0,$$

and with the two identities that follow from them:

$$Me = e, \quad MMy = My = Mu,$$

the first because $e = My$ already lies in $L(X)^\perp$, which M leaves unchanged, the second because M is idempotent and $My = M(X\beta + u) = Mu$ once $MX = 0$.

For the two blocks the page writes the matching pair,

$$P_1 \text{ projects on } L(X_1), \quad M_1 \text{ projects on } L(X_1)^\perp,$$

and records how the two spaces sit inside the full ones:

$$L(X_1) \subset L(X), \quad L(X_1)^\perp \supset L(X)^\perp.$$

The identities the page collects

$$P_1X_1 = X_1, \quad M_1X_1 = 0, \quad PX_1 = X_1, \quad M_1My = My = e,$$

with the two questions left open beside them,

$$P_1X_2 = ? \quad M_1X_2 = ?$$

The answer: in general neither is zero. P_1X_2 is the matrix of fitted values from regressing each column of X_2 on X_1 , and M_1X_2 the matching residuals; $P_1X_2 = 0$ and $M_1X_2 = X_2$ exactly when $X_1'X_2 = 0$, i.e. when the two blocks are orthogonal.

The top of the next page runs the same list for the second block,

$$P_2X_2 = X_2, \quad M_2X_2 = 0, \quad PX_2 = X_2, \quad M_2My = My, \quad P_2X_1 = ? \quad M_2X_1 = ?$$

and then writes the two matrices out in full:

$$P_1 = X_1(X_1'X_1)^{-1}X_1', \quad M_1 = I_n - P_1 = I_n - X_1(X_1'X_1)^{-1}X_1'.$$

3 The Frisch–Waugh–Lovell theorem

The theorem is headed *Frisch–Waugh–Lovell (FWL)*. For

$$y = X_1\beta_1 + X_2\beta_2 + u$$

the whole line is hit with the annihilator of the second block,

$$M_2y = M_2X_1\beta_1 + M_2u,$$

the term in β_2 dropping because $M_2X_2 = 0$. The page labels the three pieces underneath,

$$\underbrace{M_2y}_{y^*} = \underbrace{M_2X_1}_{X_1^*} \beta_1 + \underbrace{M_2u}_{u^*},$$

and the regression that uses them is the starred one,

$$(X_1^{*'}X_1^*)^{-1}X_1^{*'}y^*,$$

that is,

$$b_1 = (X_1'M_2X_1)^{-1}X_1'M_2y,$$

with the other block obtained the same way,

$$b_2 = (X_2'M_1X_2)^{-1}X_2'M_1y.$$

What the starred objects are

The page first recalls how any regression splits y ,

$$y = \underbrace{Py}_{\text{explained}} + \underbrace{My}_{\text{residuals}},$$

and then reads both starred objects as residuals of a regression on X_2 :

$$M_2y \longrightarrow \text{regressing } y \text{ on } X_2 \longrightarrow \text{residual},$$

$$M_2X_1 \longrightarrow \text{regressing } X_1 \text{ on } X_2 \longrightarrow \text{residual}.$$

Read together: $y^* = M_2y$ is what is left of y once X_2 is taken out, the regressors of the second stage are the residuals $X_1^* = M_2X_1$, and regressing the first on the second returns b_1 , the same value as in the full regression. The residuals of the starred regression are also the same as those of the full regression, and since $X_1^{*'}y^* = X_1'M_2M_2y = X_1'M_2y$, regressing y itself (rather than M_2y) on X_1^* gives the same b_1 .

The substitution, and why b_1 is unbiased

$$\begin{aligned} b_1 &= (X_1'M_2X_1)^{-1}X_1'M_2y \\ &= (X_1'M_2X_1)^{-1}X_1'M_2(X_1\beta_1 + X_2\beta_2 + u) \\ &= \beta_1 + (X_1'M_2X_1)^{-1}X_1'M_2u, \end{aligned}$$

the middle term vanishing again because $M_2X_2 = 0$.

4 The sampling distribution of the block estimators

With the substitution in hand, b_1 is unbiased and its covariance is the expectation of the outer product written on the page,

$$\mathbb{E}(b_1) = \beta_1,$$

$$\begin{aligned} \text{Var}(b_1) &= \mathbb{E}\left\{(X_1'M_2X_1)^{-1}X_1'M_2u[(X_1'M_2X_1)^{-1}X_1'M_2u]'\right\} \\ &= (X_1'M_2X_1)^{-1}X_1'M_2\mathbb{E}(uu')M_2X_1(X_1'M_2X_1)^{-1} = \sigma^2(X_1'M_2X_1)^{-1}, \end{aligned}$$

the last step using $\mathbb{E}(uu') = \sigma^2I_n$ and $M_2M_2 = M_2$. Hence

$$b_1 \sim N(\beta_1, \sigma^2(X_1'M_2X_1)^{-1}), \quad b_2 \sim N(\beta_2, \sigma^2(X_2'M_1X_2)^{-1}),$$

and, on the same line of the page, the estimator of the whole vector,

$$b \sim N(\beta, \sigma^2(X'X)^{-1}).$$

5 Misspecification: the two cases

The page turns to misspecification, listed as two cases: (1) omit relevant variables, and (2) include irrelevant variables.

6 Case (1): omitting a relevant variable

The data come from the larger model and the smaller one is estimated:

$$\underbrace{y = X_1\beta_1 + X_2\beta_2 + u}_{\text{data gen. process}}, \quad \underbrace{y = X_1\beta_1 + u}_{\text{estimated model}}$$

and the estimator of the estimated model expands as

$$\begin{aligned}\hat{\beta}_1 &= (X_1'X_1)^{-1}X_1'y \\ &= \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'u.\end{aligned}$$

Taking expectations, the second term does not disappear,

$$\mathbb{E}(\hat{\beta}_1) = \beta_1 + \underbrace{(X_1'X_1)^{-1}X_1'X_2\beta_2}_{\text{bias}},$$

which is the bias the page marks by underlining that term and writing *bias* at its arrow. The bias vanishes only if $\beta_2 = 0$ or $X_1'X_2 = 0$ (the omitted regressors are orthogonal to the included ones). The variance, by contrast, is the short-regression one, $\sigma^2(X_1'X_1)^{-1}$, not that of b_1 in the correct model:

$$\text{Var}(\hat{\beta}_1) = \mathbb{E}[(\hat{\beta}_1 - \mathbb{E}(\hat{\beta}_1))(\hat{\beta}_1 - \mathbb{E}(\hat{\beta}_1))'] = \mathbb{E}[(X_1'X_1)^{-1}X_1'uu'X_1(X_1'X_1)^{-1}] = \sigma^2(X_1'X_1)^{-1},$$

so the two sampling distributions the page puts side by side are

$$\hat{\beta}_1 \sim N(\beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2, \sigma^2(X_1'X_1)^{-1}), \quad b_1 \sim N(\beta_1, \sigma^2(X_1'M_2X_1)^{-1}).$$

The comparison of the two variances

$$\text{Var}(\hat{\beta}_1) = \sigma^2(X_1'X_1)^{-1}, \quad \text{Var}(b_1) = \sigma^2(X_1'M_2X_1)^{-1},$$

and since

$$X_1'M_2X_1 = X_1'(I_n - P_2)X_1 = X_1'X_1 - X_1'P_2X_1 \leq X_1'X_1,$$

where $\leq$ is the positive semi-definite order ($X_1'P_2X_1 = (P_2X_1)'(P_2X_1)$ is positive semi-definite), and since inverting positive definite matrices reverses that order, the inverse is at least as large,

$$(X_1'X_1)^{-1} \leq (X_1'X_1 - X_1'P_2X_1)^{-1}, \quad \text{so} \quad \text{Var}(\hat{\beta}_1) \leq \text{Var}(b_1),$$

with equality only when $X_1'X_2 = 0$: omitting a variable costs bias but buys (a smaller) variance.

Mean squared error

$$\text{MSE}(\hat{\theta}) = \mathbb{E}(\hat{\theta} - \theta)^2 = \text{Var}(\hat{\theta}) + (\text{bias})^2,$$

with the matrix version written beside it, $\mathbb{E}[(\hat{\theta} - \theta)(\hat{\theta} - \theta)']$, and applied to the estimator of this section,

$$\text{MSE}(\hat{\beta}_1) = \mathbb{E}[(\hat{\beta}_1 - \beta_1)(\hat{\beta}_1 - \beta_1)'] = (X_1'X_1)^{-1}X_1'X_2\beta_2\beta_2'X_2'X_1(X_1'X_1)^{-1} + \sigma^2(X_1'X_1)^{-1},$$

the first term being the squared bias and the second the variance. The first line of the next page gives the same quantity for the estimator of the correct model,

$$\text{MSE}(b_1) = \sigma^2(X_1' M_2 X_1)^{-1},$$

which is the value the biased estimator's mean squared error is weighed against. Neither is smaller in general: omitting X_2 lowers the MSE when the squared bias is smaller than the variance saved, e.g. when β_2 is small relative to σ .

7 Case (2): including an irrelevant variable

Now the data come from the smaller model and the larger one is estimated:

$$\underbrace{y = X_1 \beta_1 + u}_{\text{data gen. process}} \quad \underbrace{y = X_1 \beta_1 + X_2 \beta_2 + u}_{\text{estimated model}}$$

so the estimator of β_1 is the full-model one,

$$\begin{aligned} \hat{\beta}_1 &= (X_1' M_2 X_1)^{-1} X_1' M_2 y \\ &= (X_1' M_2 X_1)^{-1} X_1' M_2 (X_1 \beta_1 + u) \\ &= \beta_1 + (X_1' M_2 X_1)^{-1} X_1' M_2 u, \end{aligned}$$

unbiased,

$$\mathbb{E}(\hat{\beta}_1) = \beta_1, \quad \text{Var}(\hat{\beta}_1) = \sigma^2(X_1' M_2 X_1)^{-1},$$

but carrying the larger of the two variances: the estimator from the model that is right is

$$b_1 = (X_1' X_1)^{-1} X_1' y, \quad \mathbb{E}(b_1) = \beta_1, \quad \text{Var}(b_1) = \sigma^2(X_1' X_1)^{-1},$$

and, both of them unbiased, the comparison is between the variances; by the inequality of case (1),

$$\text{MSE}(\hat{\beta}_1) = \sigma^2(X_1' M_2 X_1)^{-1} \geq \sigma^2(X_1' X_1)^{-1} = \text{MSE}(b_1),$$

with equality only when $X_1' X_2 = 0$. Including an irrelevant variable therefore costs variance and buys nothing.

8 Inference: a linear hypothesis

The model is written once more in general form,

$$\underbrace{y}_{n \times 1} = \underbrace{X}_{n \times K} \underbrace{\beta}_{K \times 1} + \underbrace{u}_{n \times 1},$$

and the last page of the set, headed *Inference, hypothesis test*, turns to testing. The hypotheses written across the top are

$$\beta_1 = \beta_2 \quad (\text{that is } \beta_1 - \beta_2 = 0), \quad \beta_3 - \beta_5 = 1, \quad \beta_4 + \beta_6 + \beta_7 = 2,$$

and they are collected into a single linear restriction,

$$H_0 : R\beta = r, \quad \underbrace{R}_{q \times K} \underbrace{\beta}_{K \times 1} = \underbrace{r}_{q \times 1},$$

where q , in the page's own note, is the number of hypotheses (restrictions); the rows of R must be linearly independent, $\text{rank}(R) = q \leq K$, so that no restriction is redundant or contradicts another. The coefficient vector is written out as

$$\beta = (\beta_1 \ \beta_2 \ \cdots \ \beta_j \ \cdots \ \beta_K)',$$

and the three hypotheses sit in the rows of the $3 \times K$ matrix

$$R = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & \cdots & 0 \end{pmatrix}, \quad R\beta = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = C,$$

the pieces carrying the labels of the page's last line, $RB = C$: the restriction matrix, the coefficient vector B and the vector of restrictions C . The lecture set ends here, at the statement of the null hypothesis; the test statistic built on it is not written down.

Transcription notes

- The set opens with a divider page carrying only “Lesson 4”; it is not reproduced.
- The regressor matrix is written in lower case throughout the original (x, x_1, x_2); it is typeset upper case (X, X_1, X_2), as in Lesson 3. The matrices P and M are written bare on the page (the shorthand of Lesson 3) and are kept bare here, with the block versions P_1, M_1, P_2, M_2 .
- $MMy = My = Mu$ carries no subscript in the original; the same identity for the second block, on the following page, is written $M_2My = My$. Both are transcribed as written.
- The two questions $P_1X_2 = ?$ and $M_1X_2 = ?$ on the first working page are never answered on it; the next page runs the same list for the second block instead. **Added:** a short answer (fitted values and residuals of X_2 on X_1 ; zero, resp. X_2 , only when $X_1'X_2 = 0$).
- **Corrected:** the covariance of b_1 was written as the outer product itself, without the expectation; $\mathbb{E}\{\cdot\}$ is inserted and the middle step shown.
- **Added:** when the omitted-variable bias vanishes, that the FWL residuals coincide with the full-regression ones, the comparison of the two MSEs, and the rank condition $\text{rank}(R) = q$.
- Below b_2 the page writes “ My explained Py residuals” with a y under “explained”; it is the split $y = Py + My$ into explained part and residuals, and is typeset so (an earlier transcription read it as a gloss on M_2y). The next line reads “ $M_2y \rightarrow$ regression and $X_2 \rightarrow$ residual” — the subscript is 2, not 1: M_2y is the residual of regressing y on X_2 . It is typeset in the same words as the line below it, “ $M_2X_1 \rightarrow$ regressing X_1 on $X_2 \rightarrow$ residual”.
- In the substitution for b_1 the page repeats y and appends the bracketed model as an insertion; the second occurrence of the leading bracket carries a prime where the inverse belongs, and the X_1' after it is dropped. The chain is typeset once, in the corrected form, which is what the preceding line (the starred regression) and the closing line $\beta_1 + (X_1'M_2X_1)^{-1}X_1'M_2u$ both require. The closing line itself carries a stray y before M_2u ; it is dropped.
- The misspecified-model estimator is written $\hat{\beta}_1$ on the page (an earlier transcription had $\hat{b}_1$); it is typeset $\hat{\beta}_1$. The hat changes meaning between the two cases of the misspecification section: in case (1) $\hat{\beta}_1$ is the estimator from the *smaller*, mis-specified model and b_1 the estimator from the true, larger one; in case (2) $\hat{\beta}_1$ is the estimator from the *larger* model and b_1 the estimator from the true, smaller one. Both are kept as written; the text says which model each estimator comes from.

- The second line of the substitution in case (2) is written with a symbol that reads as β_1 where the estimator is meant; it is typeset $\hat{\beta}_1$.
- The first line of page 6 is written $\text{MSE}(b_1) = \sigma^2(X_1' M_2 X_1)^{-1}$ without a hat, in the sequence where the estimator of the correct model is meant; it is kept without the hat.
- **Corrected:** the inequality chain of case (1) ends $\text{Var}(\hat{\beta}_1) < \text{Var}(b_1)$ with a strict sign, and case (2) ends $\text{MSE}(\hat{\beta}_1) > \text{MSE}(b_1)$. The middle steps $X_1' X_1 \geq X_1' X_1 - X_1' P_2 X_1$ and $(X_1' X_1)^{-1} \leq (X_1' X_1 - X_1' P_2 X_1)^{-1}$ only give the weak (positive semi-definite) order, and the two are equal when $X_1' X_2 = 0$; both conclusions are typeset with $\leq / \geq$.
- **Corrected:** in case (1) the expansion of $\hat{\beta}_1$ is written $\beta_1 + (X_1' X_2)^{-1} X_1' X_2 \beta_2 + \dots$; the inverse is $(X_1' X_1)^{-1}$, as in the bias line below it. The distribution of $\hat{\beta}_1$ is written with mean $\beta_1 + (X_1' X)^{-1} X_1' X_2 \beta_1$; it is $\beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2$. The first line of $\text{MSE}(\hat{\beta}_1)$ has $\hat{\beta}_1 - \beta$ in the first factor; it is β_1 .
- **Added:** the page gives no labels to the two models of case (2); the “data gen. process” / “estimated model” braces are carried over from case (1). The general model $y = X\beta + u$ ends page 6 (dimensions written $n \times k$, $K \times 1$); it opens the inference section here, with K throughout. On page 2 the page writes “ P_1 project on $\ell(x)$ ” where $L(X_1)$ is meant, and $y = X_1\beta + X_2\beta + u$ without the block subscripts on β ; both are typeset in the intended form.
- The page heads the theorem “Frish-Waugh-Lovell”; “Frisch” is written here, as in Lesson 3.
- The column-space glyph is the cursive lower-case letter read as ℓ in Lessons 2 and 3; the spaces are typeset $L(X_1)$, $L(X)$, and the orthocomplement $L(X_1)^\perp$, $L(X)^\perp$. The page writes the full space as a bare ℓ in $\ell(X_1) \subset \ell$; it is $L(X)$.
- On the last page the coefficient vector is written as a transposed row, $\beta = (\beta_1 \ \beta_2 \ \dots \ \beta_j \ \dots \ \beta_K)'$; it is typeset as the $K \times 1$ column it denotes. The example index is a single glyph that reads as j or as 7; a generic coefficient is what the notation wants, and it is typeset β_j . Row 2 of R has +1 in the third position and -1 in the fifth, which is $\beta_3 - \beta_5 = 1$ of the headline; row 3 has entries in the fourth, sixth and seventh positions, which is $\beta_4 + \beta_6 + \beta_7 = 2$. The page writes the columns out to the eighth position and then stops; the three rows are given the same nine columns here, with $\dots$ through the middle and a closing zero, so that they line up.
- Pages of the original, for tracing back: page 1 the divider; page 2 the reminder, the block model, b and its block form, P and M with $PX = X$, $Pe = 0$, $MX = 0$, $Me = e$, $MMy = My = Mu$, the two “project” lines with the nesting of the spaces, and the identities $P_1 X_1 = X_1$, $M_1 X_1 = 0$, $PX_1 = X_1$, $M_1 My = My = e$ with the two open questions; page 3 $P_2 X_2 = X_2$, $PX_2 = X_2$, $M_2 My = My$, the two definitions, the FWL heading, $M_2 y = M_2 X_1 \beta_1 + M_2 u$ with the starred labels, the starred regression, b_1 and b_2 , the split $y = Py + My$, the two arrow lines and the substitution; page 4 $\mathbb{E}(b_1)$ and $\text{Var}(b_1)$, the three sampling distributions, the misspecification heading with its two cases, and the start of case (1); page 5 the expansion of $\text{Var}(\hat{\beta}_1)$, the two distributions, the variance comparison with its inequality chain, then $\text{MSE}(\hat{\theta})$ and $\text{MSE}(\hat{\beta}_1)$; page 6 $\text{MSE}(b_1)$, the heading of case (2), $\hat{\beta}_1$ with its expectation and variance, the comparison $\text{MSE}(\hat{\beta}_1) > \text{MSE}(b_1)$, and the general model; page 7 the three hypotheses, $H_0 : R\beta = r$ with its dimensions, the note on q , β written out, the matrix R and $RB = C$.